23AD534	UI AND UX DESIGN	L	T	P	C
		3	0	2	4
Category	Professional Core Courses				
Pre requisites	Problem Solving, HTML				

Course Objectives

The course is intended to make the students to

- Explain the basic concepts and principles of UI and UX design.
- Analyze user needs and interaction patterns for effective design.
- Apply visual design principles to create user-friendly interfaces.
- Design wireframes and prototypes for digital applications.
- Evaluate interface usability using feedback and design guidelines.

Course Outcomes

On successful completion of the course, students will be able to

CO. No	Course Outcome	Bloom's Level
Theory		
CO 1	Build UI for user Applications	Apply
CO 2	Evaluate UX design of any product or application	Apply
CO 3	Demonstrate UX Skills in product development	Apply
CO 4	Implement Sketching principles	Apply
CO 5	Create Wireframe and Prototype	Understand

Course Contents

UNIT – I	**FOUNDATIONS OF DESIGN**	**9**
UI vs. UX Design – Core Stages of Design Thinking – Divergent and Convergent Thinking – Brainstorming and Game storming – Observational Empathy.		
UNIT – II	**FOUNDATIONS OF UI DESIGN**	**9**
Visual and UI Principles – UI Elements and Patterns – Interaction Behaviors and Principles – Branding – Style Guides.		
UNIT – III	**FOUNDATIONS OF UX DESIGN**	**9**
Introduction to User Experience – Why you should Care about User Experience – Understanding User Experience – Defining the UX Design Process and its Methodology – Research in User Experience Design – Tools and Method used for Research – User Needs and its Goal – Know about Business Goals.		
UNIT – IV	**WIREFRAMING, PROTOTYING AND TESTING**	**9**
Sketching Principles – Sketching Red Routes – Responsive Design - Wire framing - Creating Wire flows - Building a Prototype - Building High-Fidelity Mock ups - Designing Efficiency with tools - Interaction Patterns - Conducting Usability Tests – Other Evaluative User Research Methods – Synthesizing Test Findings – Prototype Iteration		
UNIT – V	**RESEARCH, DESIGNING, IDEATING, & INFORMATION ARCHITECTURE**	**9**
Identifying and Writing Problem Statements – Identifying Appropriate Research Methods – Creating Personas – Solution Ideation – Creating User Stories – Creating Scenarios – Flow Diagrams – Flow Mapping – Information Architecture		
		Total: 45 Periods

List of Experiments

S.No.	**Name of the Experiment**	**CO**	**Bloom's Level**
1	Designing a Responsive layout for an societal application	CO1	Apply
2	Exploring various UI Interaction Patterns	CO1	Apply

3	Developing an interface with proper UI Style Guides	CO2	Apply
4	Developing wire flow diagram for application using open-source software	CO2	Apply
5	Exploring various open-source collaborative interface Platform	CO3	Apply
6	Hands on Design Thinking Process for a new product	CO3	Apply
7	Brainstorming feature for proposed product	CO3	Apply
8	Defining the Look and Feel of the new Project	CO4	Apply
9	Create a Sample Pattern Library for the product (Mood board, Fonts, Colors based on UI principles)	CO4	Apply
10	Identify a customer problem to solve	CO5	Apply
11	Conduct end-to-end user research - User research, creating personas, Ideation process (User stories, Scenarios), Flow diagrams, Flow Mapping	CO5	Apply
12	Sketch, design with popular tool and build a prototype and perform usability testing and identify improvements	CO5	Apply

Text Books

1. Joel Marsh, “UX for Beginners”, O’Reilly , 2022
2. Jon Yablonski, “Laws of UX using Psychology to Design Better Product & Services” O’Reilly 2021

Reference Books

1. Jenifer Tidwell, Charles Brewer, Aynne Valencia, “Designing Interface” 3 rd Edition , O’Reilly 2020

2. Steve Schoger, Adam Wathan “Refactoring UI”, 2018

3. Steve Krug, “Don’t Make Me Think, Revisited: A Commonsense Approach to Web & Mobile”, Third Edition, 2015

Additional / Web References

https://www.nngroup.com/articles/
https://www.interaction-design.org/literature

Mapping of Course Outcomes (COs) with Programme Outcomes (POs) Programme Specific Outcomes (PSOs)															
COs	**POs**												**PSOs**		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**
CO 1	3	3	2	1	-	-	-	-	-	-	-	-	2	2	2
CO 2	2	2	3	-	-	-	-	-	-	-	-	-	2	2	2
CO 3	2	2	3	-	-	-	-	-	-	-	-	-	2	2	2
CO 4	2	3	2	1	-	-	-	-	-	-	-	-	2	2	2
CO 5	3	2	1	1	1	-	-	-	-	-	-	-	2	2	2
Average	**3**	**2**	**3**	**1**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**2**	**2**

<table>
<tr><th>Assessm e nt</th><th>Duration</th><th>Syllabus to be covered</th><th>Max. Marks</th><th>Weightage for Internal Marks</th><th>Continuous Internal Assessment Marks</th><th>End Semeste Examinati on Marks</th></tr>
<tr><td colspan="7">Theory Component</td></tr>
<tr><td>CIA I</td><td>3 hours</td><td>2.5 units</td><td>100</td><td>10</td><td rowspan="2">20</td><td rowspan="2">50</td></tr>
<tr><td>CIA II</td><td>3 hours</td><td>2.5 units</td><td>100</td><td>10</td></tr>
<tr><td colspan="7">Practical Component</td></tr>
<tr><td colspan="2">Observation & Analysis of Experimental results, Viva Voce, Quiz based on rubrics.</td><td rowspan="2">All Experiments</td><td>75</td><td>22.5</td><td rowspan="2">30</td><td rowspan="2">-</td></tr>
<tr><td>Model Exam</td><td>3 hours</td><td>25</td><td>7.5</td></tr>
<tr><td colspan="5">Total</td><td>50</td><td>50</td></tr>
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